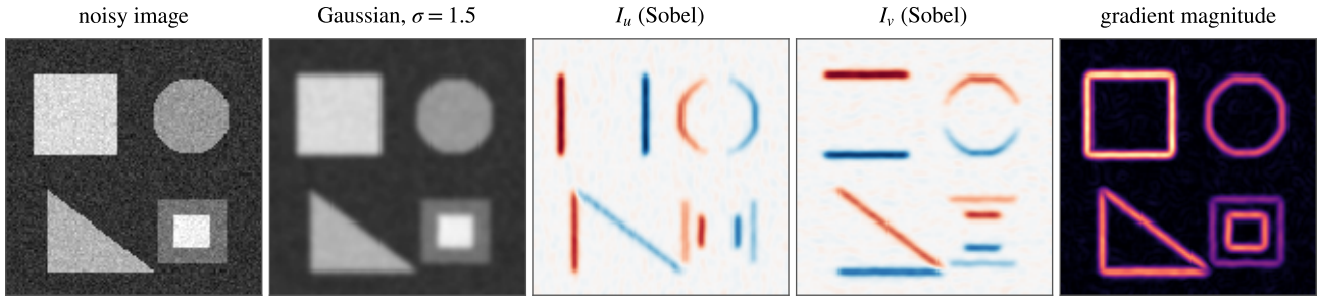

noisy image
Gaussian, $\sigma = 1.5$
I_u (Sobel)
I_v (Sobel)
gradient magnitude